

div / curl / grad / ∇?

→ Operators (Col 1)

Helix r(t)?

→ FS Chain (Col 1 btm)

Surface area?

→ Monge (Col 2 btm)

∫F·dr or work?

→ Line Int (Col 2-3)

∭ or volume?

→ Triple Int (Col 2)

∮ circulation?

→ Circ (Col 3)

SCOPE · PRIORITY · EXAM ORDER

In scope:

triple integrals, cyl/sph setup, curves, helix/FS, surface area, grad/div/curl/Laplacian, line integrals, circulation.

1. Operators

2. Helix / FS

3. Surface area

4. Line integrals

5. Triple integrals

Weak-student rule:

do not spread effort evenly. Make operator identities, helix template, Monge formula and line-integral workflow nearly automatic first.

CORE OPERATORS · CARTESIAN

∇ = 1∂x + j∂y + k∂z

∇f = (fx, fy, fz) → vector output

∇ · F = ∂xF1 + ∂yF2 + ∂zF3 → scalar output

∇ × F =

i	j	k
∂x	∂y	∂z
F1	F2	F3

 → middle term sign flips

∇²f = fxx + fyy + fzz → sum all pure 2nd derivs

Always: ∇ × (∇f) = 0, ∇ · (∇ × F) = 0.

Product rules: ∇(uv) = u∇v + v∇u, ∇ · (uF) = u∇ · F + F · ∇u, ∇ × (uF) = ∇u × F + u∇ × F

Cross-product rule: ∇ × (A × B) = A(∇ · B) − B(∇ · A) + (B · ∇)A − (A · ∇)B

If you see → scalar input: use ∇ or ∇². Vector input: use ∇ · or ∇ ×. Check output type matches.

RADIAL IDENTITIES · MUST MEMORIZE

r = xi + yj + zk, r = |r| = √(x² + y² + z²)

∇ · r = 3 → instant, ∇ × r = 0 → instant, ∇(r²) = 2r → instant

∇(rⁿ) = nrⁿ⁻²r = nrⁿ⁻¹ê, ∇(1/r) = −r/r³ = −ê/r²

Radial Laplacian: ∇²f(r) = f''(r) + 2/r f'(r) → critical for ln r, rⁿ types

If you see (x² + y² + z²)ⁿ or h(r) → convert to radial form rᵏê FIRST, then apply radial formula.

2024 auto-marks: ∇ · r = 3, ∇ × r = 0, ∇r² = 2r. Write these instantly.

Common trap: mixing r and r², or writing scalar when vector is needed → fix: check output type after every computation.

CURVES · ARCLENGTH · HELIX

r(t) = (x(t), y(t), z(t)), v = r'(t), ds = |r'(t)|dt, s(t) = ∫t₀ |r'(u)| du

Helix: r(t) = (a cos t, a sin t, bt), r' = (−a sin t, a cos t, b), |r'| = c = √(a² + b²) → constant

If you see r(t) = (a cos t, a sin t, bt) → it is a helix. Execute the FS chain below.

Exam sequence: differentiate → speed → arclength → unit tangent → unit normal → binormal → curvature/torsion.

FRENET-SERRET CORE

ê = dx/ds = r'/|r'| → normalize first

κ = |dê/ds|

dê/ds = κê⊥ → then â = 1/κ dê/ds

b̂ = ê × â → cross product order matters

db̂/ds = −τâ

Helix results: κ = a/(a² + b²), τ = b/(a² + b²), Ω = κb̂ + τê

Common trap: computing dê/ds and calling it κ → fix: ALWAYS divide by |r'| to convert ds → ds

CYLINDRICAL / SPHERICAL OPERATORS

Cyl: ∇f = frê + 1/r fφê + fzê

∇ · F = 1/r ∂r(rFr) + 1/r ∂θFθ + ∂zFz → note 1/r ∂r(rFr) structure

∇²f = 1/r ∂r(rfr) + 1/r² fθθ + fzz

Sph: ∇f = frê + 1/r fφê + 1/r sin φ fθê

∇ · F = 1/r² ∂r(r²Fr) + 1/r sin φ ∂φ(sin φ Fφ) + 1/r sin φ ∂θFθ

∇²f = 1/r² ∂r(r²fr) + 1/r² sin φ ∂φ(sin φ fφ) + 1/r² sin² φ fθθ → use if f depends on φ or θ

Purely radial F = Fr(r)ê: ∇ · F = 1/r² d/dr(r²Fr), ∇ × F = 0 away from singularities.

If you see (x² + y² + z²)ⁿ(x, y, z) → rewrite as r²ⁿ⁺¹ê, identify Fr, use spherical div formula.

Common trap: forgetting ê = r/r introduces extra r → fix: write Fr explicitly before applying formula.

COORDINATES · JACOBIANS · BOUNDS

Polar / Cyl: x = r cos θ, y = r sin θ, dA = r dr dθ, dV = r dr dθ dz

Spherical: x = r sin φ cos θ, y = r sin φ sin θ, z = r cos φ

dV = r² sin φ dr dφ dθ → never drop sin φ

First octant ⇒ 0 ≤ θ ≤ π/2, 0 ≤ φ ≤ π/2

Cone z = r ⇒ φ = π/4

Sphere radius a ⇒ r = a

Offset cylinder x² + (y − 1)² = 1 ⇒ r = 2 sin θ

Setup rule: sketch the shadow region first, then choose coordinates by the geometry trigger table.

TRIPLE-INTEGRAL MICRO-TEMPLATES

Volume: V(D) = ∭D 1 dV → integrand is 1, not the surface

Centroid: z̄ = 1/V ∭D z dV → divide by V at the end

Cartesian: inner variable runs from lower surface to upper surface. → sketch region first

Cylindrical: sidewall → r bound. Floor → z lower. Roof → z upper. θ from geometry. Attach Jacobian r.

Spherical: sphere → r bound. Cone → φ bound. Octant → θ bound. Attach r² sin φ.

If you see ∭ → sketch solid, pick coordinates by geometry, write bounds, attach Jacobian.

EP2 Q11 pattern: sidewall r = 2 sin θ, floor z = 0, roof z = r, dV = r dz dr dθ

Common trap: writing dV = dr dθ dz without Jacobian → fix: cyl → r, sph → r² sin φ

SURFACE PARAMETRIZATION · AREA

Parametric surface: X(u, v)

dS = |Xu × Xv| du dv

Graph surface: z = f(x, y)

X(x, y) = (x, y, f(x, y))

dS = √(1 + f²x + f²y) dx dy → do not forget the √

Always find the projection region in the xy-plane before integrating. → sketch region

Paraboloid cap template: solve intersection first. If z = 2 − r² and cone z = r, then 2 − r² = r ⇒ r = 1

Then A = ∫₀² ∫₀¹ √(1 + 4r²) r dr dθ

If you see "find the area of z = f(x, y)" → Monge formula: √(1 + f²x + f²y), project onto xy-plane.

Common trap: using the bounding surface's derivatives instead of the named surface → fix: check which surface the question asks for.

COMMON SURFACE-AREA PATTERNS

Problem type	Minimal skeleton
Graph z = f(x, y)	fx, fy → √(1 + f²x + f²y) → project region → polar if circular
Sphere	parameterize by angles → Xu × Xv
Paraboloid above cone	solve intersection in r first, then Monge formula on paraboloid

LINE INTEGRALS · DIRECT METHOD

∫C F · dx = ∫aᵇ F(r(t)) · r'(t) dt → substitute path into F first

In 2D: ∫C P dx + Q dy

Workflow: parameterize C → compute dx → write F(r(t)) explicitly → dot → integrate.

LINE-INTEGRAL PATTERNS TO DRILL

Type	Minimal symbolic template
Straight segment	r(t) = A + t(B − A), 0 ≤ t ≤ 1
Circle	r(t) = (a cos t, a sin t) or 3D variant
Helix	r(t) = (cos t, sin t, t), r' = (−sin t, cos t, 1)
Broken path	split C = C1 ∪ C2 ∪ ...; integrate each, sum
Path in 2D	substitute y = y(x), dy = y'(x)dx, single-variable integral

2023 MT2 pattern: compute work on a helix, then on a straight segment between the same endpoints and compare. → values will differ

CIRCULATION TEMPLATE · §4.4

Circulation = ∫C F · dx net tangential component of F around a closed curve. → CCW = positive

Step 1. Parameterize: circle radius a → r(t) = (a cos t, a sin t), 0 ≤ t ≤ 2π → CCW

Step 2. r'(t) = (−a sin t, a cos t)

Step 3. Substitute: F(r(t)) · r'(t) → simplify → integrate over [0, 2π].

If you see f or "circulation" → parameterize closed curve (CCW), compute r', substitute F, dot, integrate.

Common trap: wrong orientation reverses sign → fix: confirm CCW. Reversing direction flips the answer.

TOP 10 EXAM-READY DRILLS

#	Source	Why it matters
1	2025 MT2 Q2(a)–(b)	Best complete radial-field divergence/curl drill
2	2024 MT2 Q1(a)–(c)	Fast operator identities, nearly automatic marks
3	Tutorial 7 Q1(a)–(c)	Derive radial-gradient identities, not just memorize
4	2022 MT2 Q1(b)	Best Frenet-Serret frame drill
5	2025 MT2 Q1	Best graph-surface area exam problem
6	2023 MT2 Q2(a)	Best 3D direct line-integral drill
7	2023 MT2 Q1(c)	Radial divergence with parameter condition
8	Tutorial 7 Q2	Training version of paraboloid-area setup
9	2022 MT2 Q1(c)	Curvature and torsion follow-up after frame mastery
10	Extra Problems 2 Q11(a)	Best cylindrical triple-integral setup drill

COMMON BOUND / GEOMETRY TRIGGERS

Given geometry	Immediate move
x² + y² everywhere	Switch to polar/cylindrical
x² + y² + z² or sphere/cone	Test spherical first
Graph z = f(x, y)	Project onto xy-plane, use Monge
Patch inside cylinder	Cylinder defines boundary disk only
Offset circle/cylinder	Rewrite in polar: r = function of θ
First octant	All coords ≥ 0, angle bounds shrink
Helix	Use parametric t, speed is constant

COMMON MISTAKES TO ELIMINATE

Operators: writing scalar where vector is required, or vice versa. → check output type

Radial questions: failing to rewrite field in radial form before differentiating. → convert first

Frenet-Serret: using d/dt as if it were d/ds. → divide by |r'|

Surface area: forgetting projection region or using wrong surface's derivatives. → name the surface

Triple integrals: missing Jacobian r or r² sin φ → write Jacobian immediately

Line integrals: not substituting both path and differential. → write F(r(t)) first

Circulation: wrong orientation reverses sign. → confirm CCW

Exam habit: before integrating, say silently: formula, bounds, Jacobian, orientation, variable order.

ONE-LINE METHOD CHECKLISTS

Operator problem: Write formula first → identify operator → differentiate componentwise → simplify.

Helix problem: Write r' first → |r'| → s → ê → â → b̂ → κ, τ.

Surface-area problem: Write dS first → choose graph/parametrization → project region → integrate.

Triple-integral problem: Sketch solid first → choose coordinates → write bounds → attach Jacobian.

Line integral: Parameterize C first → compute dx → substitute F(r(t)) · r'(t) → integrate.

Page 1 of 2. Strict scope only: Ch. 2 §2.4 → Ch. 4 §4.4.

Goal: automatic operator marks + helix workflow + area setup + direct line-integral workflow.

OPERATOR TEMPLATE A · INSTANT IDENTITIES

1. $\nabla \cdot \mathbf{r} = 1 + 1 + 1 = 3$ 2. $\nabla \times \mathbf{r} = \mathbf{0}$ (all cross-partials cancel) 3. $\nabla r^2 = 2\mathbf{r}$
Derivation of 3: $\nabla(x^2 + y^2 + z^2) = (2x, 2y, 2z) = 2\mathbf{r}$ → write these three instantly for free marks
Use on: 2024 MT2 Q1(a)–(c), Tutorial 7 Q1(a), Lecture 3A operator warm-ups.

OPERATOR TEMPLATE B · RADIAL FIELD

If $\mathbf{F} = (x^2 + y^2 + z^2)^n(x, y, z)$, then
 $\mathbf{F} = r^{2n}\mathbf{r} = r^{2n+1}\hat{\mathbf{r}}$ → identify $\mathbf{F}_{\mathbf{r}} = r^{2n+1}$
Divergence route:
 $\nabla \cdot \mathbf{F} = \frac{1}{r^2} \frac{d}{dr}(r^2 F_{\mathbf{r}}) = \frac{1}{r^2} \frac{d}{dr}(r^{2n+3}) = (2n+3)r^{2n}$
Solenoidal condition: $(2n+3)r^{2n} = 0 \Rightarrow n = -\frac{3}{2}$ (away from singularities).
Curl route: pure radial field $\Rightarrow \nabla \times \mathbf{F} = \mathbf{0}$ away from singularities. → state this directly
Common trap: losing the extra r when converting $\mathbf{r}$ to $\hat{\mathbf{r}}$ → fix: $\mathbf{r} = r\hat{\mathbf{r}}$, so $r^{2n}\mathbf{r} = r^{2n+1}\hat{\mathbf{r}}$.

OPERATOR TEMPLATE C · $\nabla(r^n)$ AND $\nabla(1/r)$

$r^n = (x^2 + y^2 + z^2)^{n/2}$
 $\partial_x(r^n) = \frac{n}{2}(x^2 + y^2 + z^2)^{n/2-1}(2x) = nxr^{n-2}$
Similarly, $\partial_y(r^n) = ny r^{n-2}$, $\partial_z(r^n) = nz r^{n-2}$ → same pattern for all
So $\nabla(r^n) = nr^{n-2}(x, y, z) = nr^{n-2}\mathbf{r}$
Set $n = -1$: $\nabla(1/r) = -r^{-3}\mathbf{r} = -\hat{\mathbf{r}}/r^2$
Use on: Tutorial 7 Q1(b)–(c), Lecture 3A Example 3.17, radial simplifications in past papers.

OPERATOR TEMPLATE D · $\nabla \times (\mathbf{k} \times \mathbf{r})$ [2023 Q1(a)]

Identity: $\nabla \times (\mathbf{A} \times \mathbf{B}) = \mathbf{A}(\nabla \cdot \mathbf{B}) - \mathbf{B}(\nabla \cdot \mathbf{A}) + (\mathbf{B} \cdot \nabla)\mathbf{A} - (\mathbf{A} \cdot \nabla)\mathbf{B}$
Step 1. $\mathbf{k}$ constant $\Rightarrow \nabla \cdot \mathbf{k} = 0$, $(\mathbf{r} \cdot \nabla)\mathbf{k} = \mathbf{0}$
Step 2. $(\mathbf{k} \cdot \nabla)\mathbf{r} = \mathbf{k}$ → since $\partial_i r_j = \delta_{ij}$
Step 3. $\nabla \times (\mathbf{k} \times \mathbf{r}) = \mathbf{k}(\nabla \cdot \mathbf{r}) - \mathbf{0} + \mathbf{0} - \mathbf{k} = 3\mathbf{k} - \mathbf{k} = 2\mathbf{k}$
General rule: $\nabla \times (\mathbf{c} \times \mathbf{r}) = 2\mathbf{c}$ for ANY constant vector $\mathbf{c}$. If you see $\nabla \times (\mathbf{k} \times \mathbf{r})$ → answer is $2\mathbf{k}$ instantly.

OPERATOR TEMPLATE E · LAPLACIAN OF $\ln(r^2)$ [2023 Q1(b)]

$\ln(x^2 + y^2 + z^2) = \ln(r^2) = 2\ln r$. Set $f(r) = 2\ln r$.
Step 1. $f'(r) = 2/r$, $f''(r) = -2/r^2$
Step 2. $\nabla^2 f = f''(r) + \frac{2}{r} f'(r) = -\frac{2}{r^2} + \frac{4}{r^2} = \frac{2}{r^2}$
If you see $\nabla^2(\ln(x^2 + y^2 + z^2))$ → rewrite as $2\ln r$, apply radial Laplacian, answer is $2/r^2$.

OPERATOR TEMPLATE F · PRODUCT-RULE APPLICATION

Rules: $\nabla \cdot (\mathbf{u}\mathbf{F}) = \mathbf{u}\nabla \cdot \mathbf{F} + \mathbf{F} \cdot \nabla \mathbf{u}$, $\nabla \times (\mathbf{u}\mathbf{F}) = \nabla \mathbf{u} \times \mathbf{F} + \mathbf{u} \nabla \times \mathbf{F}$
Example: $\nabla \cdot (r^2\mathbf{r})$. Set $\mathbf{u} = r^2$, $\mathbf{F} = \mathbf{r}$.
Step 1. $\nabla \mathbf{u} = \nabla(r^2) = 2\mathbf{r}$, $\nabla \cdot \mathbf{r} = 3$ → instant facts
Step 2. $\nabla \cdot (r^2\mathbf{r}) = r^2(3) + \mathbf{r} \cdot (2\mathbf{r}) = 3r^2 + 2r^2 = 5r^2$
If you see $\nabla \cdot (f\mathbf{r})$ or $\nabla \times (f\mathbf{r})$ → apply product rule, then use instant identities for each piece.

HELIX TEMPLATE A · COMPLETE FS CHAIN + DARBOUX

Step 1. $\mathbf{r}(t) = (a \cos t, a \sin t, bt)$. $\mathbf{r}'(t) = (-a \sin t, a \cos t, b)$
Step 2. $|\mathbf{r}'| = c = \sqrt{a^2 + b^2}$ → constant
Step 3. $\hat{\mathbf{t}} = \frac{1}{c}(-a \sin t, a \cos t, b)$ → normalize
Step 4. $\frac{d\hat{\mathbf{t}}}{dt} = \frac{1}{c}(-a \cos t, -a \sin t, 0)$
Step 5. $\frac{d\hat{\mathbf{t}}}{ds} = \frac{1}{c^2}(-a \cos t, -a \sin t, 0)$ → divide by c
Step 6. $\kappa = \frac{a}{c^2} \cdot \hat{\mathbf{n}} = (-\cos t, -\sin t, 0)$
Step 7. $\hat{\mathbf{b}} = \hat{\mathbf{t}} \times \hat{\mathbf{n}} = \frac{1}{c}(b \sin t, -b \cos t, a)$
Step 8. Torsion: $\frac{d\hat{\mathbf{b}}}{ds} = \frac{1}{c^2}(b \cos t, b \sin t, 0) = -\tau \hat{\mathbf{n}} \rightarrow \tau = \frac{b}{c^2}$. Darboux: $\mathbf{\Omega} = \kappa \hat{\mathbf{b}} + \tau \hat{\mathbf{t}}$
Memorize: $\kappa = a/c^2$, $\tau = b/c^2$ where $c = \sqrt{a^2 + b^2}$. Darboux: substitute $\kappa, \tau, \hat{\mathbf{b}}, \hat{\mathbf{t}}$ from above.
Numerical check ($a=3, b=4$): $c = 5$, $\hat{\mathbf{t}} = \frac{1}{5}(-3 \sin t, 3 \cos t, 4)$, $\kappa = 3/25$, $\tau = 4/25$

SURFACE-AREA TEMPLATE A · GRAPH SURFACE

If $z = f(x, y)$, then $A = \iint_R \sqrt{1 + f_x^2 + f_y^2} dA$ → Monge formula
Paraboloid: $f(x, y) = x^2 + y^2$
Step 1. $f_x = 2x$, $f_y = 2y$
Step 2. $\sqrt{1 + 4x^2 + 4y^2} = \sqrt{1 + 4r^2}$ → convert to polar
Step 3. If $0 \leq z \leq 2$, then $x^2 + y^2 \leq 2$, so $r \leq \sqrt{2}$ → projection region
Step 4. $A = \int_0^{2\pi} \int_0^{\sqrt{2}} \sqrt{1 + 4r^2} r dr d\theta$
Use on: Tutorial 7 Q2 and as training for 2025 MT2 Q1.

SURFACE-AREA TEMPLATE B · 2025 CAP

Step 1. Paraboloid: $z = 2 - r^2$. Cone: $z = r$.
Step 2. Intersection: $2 - r^2 = r \Rightarrow r^2 + r - 2 = 0 \Rightarrow r = 1$ → find intersection first
Step 3. $f_x = -2x$, $f_y = -2y$ → derivatives of the paraboloid
Step 4. $\sqrt{1 + 4x^2 + 4y^2} = \sqrt{1 + 4r^2}$
Step 5. $A = \int_0^{2\pi} \int_0^1 \sqrt{1 + 4r^2} r dr d\theta$
Common trap: using cone derivatives instead of paraboloid derivatives → fix: the area is of the paraboloid cap, not the cone.

LINE-INTEGRAL TEMPLATE A · HELIX

Step 1. $\mathbf{r}(t) = (\cos t, \sin t, t)$. $\mathbf{r}'(t) = (-\sin t, \cos t, 1)$
Step 2. For $\mathbf{F} = (-y, x, z)$: $\mathbf{F}(\mathbf{r}(t)) = (-\sin t, \cos t, t)$ → substitute path into $\mathbf{F}$
Step 3. $\mathbf{F} \cdot \mathbf{r}' = \sin^2 t + \cos^2 t + t = 1 + t$ → simplifies!
Step 4. $W = \int_0^b (1 + t) dt$
Lesson: after substitution, ugly 3D integrals often collapse to simple scalar integrands.

LINE-INTEGRAL TEMPLATE B · STRAIGHT SEGMENT

Step 1. From A to B : $\mathbf{r}(t) = A + t(B - A)$, $0 \leq t \leq 1$
Step 2. $d\mathbf{x} = (B - A) dt$ → constant direction
Step 3. $\int_C \mathbf{F} \cdot d\mathbf{x} = \int_0^1 \mathbf{F}(\mathbf{r}(t)) \cdot (B - A) dt$
Common trap: assuming same endpoints $\Rightarrow$ same value → fix: without conservative-field theorem (out of scope), different paths give different values.

LINE-INTEGRAL TEMPLATE C · 2D DIFFERENTIAL FORM

$\int_C P dx + Q dy$: substitute $y = y(x)$, $dy = y'(x)dx \rightarrow \int_a^b (P + Qy')dx$ → single variable
Broken path: integrate each segment separately and add. Circle: $x = a \cos t$, $y = a \sin t$ → check orientation

CIRCULATION TEMPLATE · WORKED EXAMPLE

Example: $\mathbf{F} = (-y, x)$. C : unit circle CCW.
Step 1. $\mathbf{r}(t) = (\cos t, \sin t)$, $0 \leq t \leq 2\pi$. $\mathbf{r}' = (-\sin t, \cos t)$
Step 2. $\mathbf{F}(\mathbf{r}(t)) = (-\sin t, \cos t)$
Step 3. $\mathbf{F} \cdot \mathbf{r}' = \sin^2 t + \cos^2 t = 1$
Step 4. $\oint = \int_0^{2\pi} 1 dt = 2\pi$
Lesson: after substitution, circulation integrals often simplify to constants.

TRIPLE-INTEGRAL TEMPLATE A · NESTED CARTESIAN

$\iiint f(x, y, z) dz dx dy$
Rule: integrate inside first, substitute its bounds completely, then move outward. → inside out
Never: carry the wrong variable into the next stage after evaluating bounds.

TRIPLE-INTEGRAL TEMPLATE B · CYLINDRICAL REGION

Step 1. Region: inside $x^2 + (y - 1)^2 = 1$, above $z = 0$, below $z = \sqrt{x^2 + y^2}$
Step 2. Sidewall: $x^2 + y^2 - 2y = 0 \Rightarrow r = 2 \sin \theta$ → sidewall gives r bound
Step 3. Floor $z = 0$, roof $z = r$ → floor/roof give z bounds
Step 4. $0 \leq \theta \leq \pi$, $0 \leq r \leq 2 \sin \theta$, $0 \leq z \leq r$
Step 5. $V = \int_0^\pi \int_0^{2 \sin \theta} \int_0^r 1 \cdot r dz dr d\theta$ → the r is the Jacobian
 $\bar{z} = \frac{1}{V} \int_0^\pi \int_0^{2 \sin \theta} \int_0^r z r dz dr d\theta$ → divide by V at end
Common trap: the extra r in the integrand is the Jacobian, not from the roof $z = r$ → fix: write Jacobian separately.

TRIPLE-INTEGRAL TEMPLATE C · SPHERICAL BOUNDS

Sphere: $\mathbf{r} = a$
Cone: $z = r \Rightarrow \phi = \pi/4$
First octant: $0 \leq \theta \leq \pi/2$, $0 \leq \phi \leq \pi/2$
 $dV = r^2 \sin \phi dr d\phi d\theta$ → never drop $r^2 \sin \phi$
Use on: Extra Problems 2 Q8 and any sphere/cone bound-construction question.

LAST-48-HOUR PRACTICE ORDER		
Block	Problems	Purpose
1	2024 MT2 Q1(a)–(c), 2023 MT2 Q1(a)–(b), Tut 7 Q1	Operators + cross-product + Laplacian
2	2025 MT2 Q2(a)–(b), 2023 MT2 Q1(c)	Radial-field recognition
3	2022 MT2 Q1(a)–(c)	Helix + FS chain
4	Tut 7 Q2, 2025 MT2 Q1	Graph-surface area
5	2023 MT2 Q2(a)–(b), Tut 8 Q1	Line integrals / path comparison
6	EP2 Q8, Q11(a)	3D bounds and Jacobians

30-SECOND PRE-SUBMISSION CHECK

1. Did I choose the correct formula/operator first?
2. Are my bounds matched to the picture and coordinate system?
3. Did I include the Jacobian r or $r^2 \sin \phi$?
4. For FS, did I convert dj/dt to d/ds ?
5. For line integrals, did I substitute both the path and $d\mathbf{x}$?
6. For circulation, is my orientation correct (CCW = positive)?
7. Did I accidentally use any out-of-scope theorem shortcut?